

Parul[®] University
Vadodara, Gujarat

NAAC
GRADE **A++**

FACULTY OF ENGINEERING AND
TECHNOLOGY BACHELOR OF TECHNOLOGY

MOBILE APP DEVELOPMENT LABORATORY
(303105380)

VI SEMESTER

CSE Department



Laboratory Manual
Session 2025-26

CERTIFICATE

This is to certify that

Mr / Ms _____ with enrollment no.

_____ has successfully completed
his/her laboratory experiments in the **Mobile App Development**
(303105380) from the department of **COMPUTER SCIENCE &
ENGINEERING** during the academic year 2025-2026.



Date of Submission:.....

Staff In charge:.....

Head Of Department:.....



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Sr. No	Experiment Title	Page No		Date of Start	Date of Completion	Marks (out of 10)	Sign
		From	To				
1.	Create a "Hello World" application: Display "Hello World" at the center of the screen, both on the Android emulator and an actual Android device.						
2.	Build an app to showcase Android lifecycle phases: Develop an app that demonstrates various Android lifecycle stages (onCreate, onStart, onResume, etc.).						
3.	Create an app with two activities: The first activity should contain an EditText and a "Send" button. When the button is clicked, use an explicit intent to send the text from EditText to a second activity and display it in a TextView						
4.	Create an app with explicit intent: The first activity should have an EditText and a "Send" button. On button click, use an implicit intent with the "SEND" action, allowing the user to select an app from an app chooser to handle the intent and display the text.						
5.	Build a basic calculator app: Create an app that performs basic arithmetic operations (addition, subtraction, multiplication, and division) on numbers						
6.	Create a Spinner-based app: Develop an app with a spinner populated from the res/values/strings.xml resource. When the spinner value changes, the corresponding image from the res/drawable directory should be displayed						
7.	Create a discount calculator app: Use a RadioGroup with three radio buttons for 10%, 15%, and 20% discounts on a shopping bill. The user can enter the bill amount in an EditText, and the selected discount will be calculated and displayed in a TextView.						
8.	Create an app with a course selection RadioButton group: Display a list of college courses with a RadioButtongroup. When a course is selected, the corresponding TIC (Total Instructional Credit) should be shown in a TextView.						

9.	Create a shopping list app using checkboxes: Build an app with checkboxes for shopping list items. As items are checked off, the selected items should be displayed in a TextView.						
10.	Create a login and registration app: Develop a login application that verifies the username and password. Include a registration page for new users. Upon successful login, show a "Welcome User" pop-up message						
11.	Create a login app with navigation to another activity: The login screen should verify the username and password. After successful login, navigate to a new activity that displays a "Welcome User" message in a TextView and a "Logout" button. On clicking "Logout," show a confirmation dialog with "OK" and "Cancel" buttons. "OK" should return to the login screen, while "Cancel" should keep the user on the current activity.						
12.	Create an app with a menu: Implement a menu with five options. The selected option should be displayed in a TextView.						
13.	Build an app using LinearLayout: Create a simple app that uses LinearLayout. It should take the contents of a predefined TextView, convert it to uppercase on button click, and display it in an EditText. Additionally, create an app that responds to key events in the EditText without needing a button press						
14.	Create an app with TableLayout and custom styles: Use a TableLayout with a TextView, EditText, and buttons. Also, create a custom styles.xml in the res/values directory to style the TextView.						
15.	Create an app with SQLite database operations: Build an app that allows the user to perform CRUD operations (Create, Read, Update, Delete) with an SQLite database						
16.	Create an app with three vertically aligned buttons: Develop an app with three buttons arranged vertically. When any button is selected, the screen color should change accordingly.						



PRACTICAL NO: 06

AIM: To develop an Android application using a **Spinner** populated from **res/values/strings.xml** that changes the displayed image based on the selected item from the spinner. The corresponding image will be loaded from the **res/drawable** directory whenever the user selects a new value.

CODE SNIPPET:

strings.xml

```
<resources>
<string name="app_name">SpinnerImageApp</string>

<string-array name="animal_names">
<item>Dog</item>
<item>Cat</item>
<item>Lion</item>
<item>Tiger</item>
</string-array>
</resources>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <Spinner
        android:id="@+id/spinnerAnimals"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />

    <ImageView
        android:id="@+id/imageViewAnimal"
        android:layout_width="250dp"
        android:layout_height="250dp"
        android:layout_marginTop="30dp"
        android:contentDescription="Animal Image" />
```

</LinearLayout>

Add images in res/drawable

Save these images with exact names:

- **dog.png**
- **cat.png**
- **lion.png**
- **tiger.png**

MainActivity.java

```
package com.example.spinnerimageapp;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

import android.view.View;

import android.widget.AdapterView;

import android.widget.ArrayAdapter;

import android.widget.ImageView;

import android.widget.Spinner;

public class MainActivity extends AppCompatActivity {

    Spinner spinner;

    ImageView imageView;

    String[] animalNames = {"Dog", "Cat", "Lion", "Tiger"};

    int[] animalImages = {R.drawable.dog, R.drawable.cat, R.drawable.lion,
R.drawable.tiger};

    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        spinner = findViewById(R.id.spinnerAnimals);
```

```
imageView = findViewById(R.id.imageViewAnimal);
```

```
ArrayAdapter<CharSequence> adapter = ArrayAdapter.createFromResource(  
    this,  
    R.array.animal_names,  
    android.R.layout.simple_spinner_item  
);
```

```
adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_it  
em);
```

```
spinner.setAdapter(adapter);
```

```
spinner.setOnItemClickListener(new  
AdapterView.OnItemClickListener() {
```

```
    @Override
```

```
    public void onItemClick(AdapterView<?> parent, View view, int  
position, long id) {
```

```
        imageView.setImageResource(animalImages[position]);
```

```
    }
```

```
    @Override
```

```
    public void onNothingSelected(AdapterView<?> parent) {
```

```
    }
```

```
});
```

```
}
```

```
}
```

OUTPUT:



PRACTICAL NO: 07

AIM: To develop an Android application where the user enters a bill amount, selects a discount using RadioButtons (10%, 15%, 20%) and the app calculates and displays: Discount Amount, Final Payable Amount.

CODE SNIPPET:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/edtBill"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Bill Amount"
        android:inputType="numberDecimal" />

    <RadioGroup
        android:id="@+id/radioGroupDiscount"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:layout_marginTop="20dp">

        <RadioButton
            android:id="@+id/radio10"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="10% Discount" />

        <RadioButton
            android:id="@+id/radio15"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="15% Discount" />

        <RadioButton
            android:id="@+id/radio20"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="20% Discount" />
    </RadioGroup>
```

```
<TextView
android:id="@+id/txtResult"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Discounted Amount Will Appear Here"
android:textSize="18sp"
android:layout_marginTop="30dp" />

</LinearLayout>
```

MainActivity.java

```
package com.example.discountcalculator;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.EditText;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    EditText edtBill;
    RadioGroup radioGroup;
    TextView txtResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        edtBill = findViewById(R.id.edtBill);
        radioGroup = findViewById(R.id.radioGroupDiscount);
        txtResult = findViewById(R.id.txtResult);

        radioGroup.setOnCheckedChangeListener((group, checkedId) -> calculateDiscount());
    }

    private void calculateDiscount() {
        String billText = edtBill.getText().toString();

        if (billText.isEmpty()) {
            txtResult.setText("Please enter bill amount!");
            return;
        }

        double bill = Double.parseDouble(billText);
        double discountPercent = 0;

        int selectedId = radioGroup.getCheckedRadioButtonId();
        if (selectedId == R.id.radio10) {
```

```
discountPercent = 10;
} else if (selectedId == R.id.radio15) {
discountPercent = 15;
} else if (selectedId == R.id.radio20) {
discountPercent = 20;
}

double discountAmount = (bill * discountPercent) / 100;
double finalAmount = bill - discountAmount;

txtResult.setText("Discount: ₹" + discountAmount + "\nFinal Bill: ₹" + finalAmount);
}
}
```

OUTPUT:



PRACTICAL NO: 08

AIM: To develop an Android application where the user selects a **college course** from a, RadioButton group and the app displays the **TIC (Total Instructional Credit)** for the selected course in a **TextView**.

CODE SNIPPET:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <RadioGroup
        android:id="@+id/radioGroupCourse"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical">

        <RadioButton
            android:id="@+id/radioBCA"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="BCA" />

        <RadioButton
            android:id="@+id/radioBBA"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="BBA" />

        <RadioButton
            android:id="@+id/radioMCA"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="MCA" />

        <RadioButton
            android:id="@+id/radioMBA"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="MBA" />
    </RadioGroup>
```

```
<TextView
android:id="@+id/txtTic"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Select a course to view TIC"
android:textSize="18sp"
android:layout_marginTop="30dp" />

</LinearLayout>
```

MainActivity.java

```
package com.example.courseselection;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.RadioGroup;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    RadioGroup radioGroup;
    TextView txtTic;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        radioGroup = findViewById(R.id.radioGroupCourse);
        txtTic = findViewById(R.id.txtTic);

        radioGroup.setOnCheckedChangeListener((group, checkedId) -> {

            int ticValue = 0;

            switch (checkedId) {
                case R.id.radioBCA:
                    ticValue = 120;
                    break;

                case R.id.radioBBA:
                    ticValue = 110;
                    break;

                case R.id.radioMCA:
                    ticValue = 150;
                    break;

                case R.id.radioMBA:
                    ticValue = 160;
                    break;
            }
        })
    }
}
```

```
txtTic.setText("TIC: " + ticValue);  
});  
}  
}
```

OUTPUT:



PRACTICAL NO: 09

AIM: To develop an Android app that uses **CheckBoxes** for a shopping list. When a user selects (checks) or deselects an item, the list of selected items is displayed in a **TextView**.

CODE SNIPPET:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent">
```

```
    <LinearLayout
```

```
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
```

```
        android:orientation="vertical"
```

```
        android:padding="20dp">
```

```
        <CheckBox
```

```
            android:id="@+id/chkMilk"
```

```
            android:layout_width="wrap_content"
```

```
            android:layout_height="wrap_content"
```

```
            android:text="Milk" />
```

```
        <CheckBox
```

```
            android:id="@+id/chkBread"
```

```
            android:layout_width="wrap_content"
```

```
            android:layout_height="wrap_content"
```

```
            android:text="Bread" />
```

```
        <CheckBox
```

```
            android:id="@+id/chkEggs"
```

```
            android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"
```

```
android:text="Eggs" />
```

```
<CheckBox
```

```
    android:id="@+id/chkRice"
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
    android:text="Rice" />
```

```
<CheckBo
```

```
    android:id="@+id/chkOil"
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
    android:text="Oil" />
```

```
<TextView
```

```
    android:id="@+id/txtSelected"
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
    android:textSize="18sp"
```

```
    android:layout_marginTop="30dp"
```

```
    android:text="Selected Items Will Appear Here" />
```

```
</LinearLayout>
```

```
</ScrollView>
```


MainActivity.java

```
package com.example.shoppinglist;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.os.Bundle;
```

```
import android.widget.CheckBox;
```

```
import android.widget.CompoundButton;
```

```
import android.widget.TextView;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    CheckBox chkMilk, chkBread, chkEggs, chkRice, chkOil;
```

```
    TextView txtSelected;
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);
```

```
        chkMilk = findViewById(R.id.chkMilk);
```

```
        chkBread = findViewById(R.id.chkBread);
```

```
        chkEggs = findViewById(R.id.chkEggs);
```

```
        chkRice = findViewById(R.id.chkRice);
```

```
        chkOil = findViewById(R.id.chkOil);
```

```
        txtSelected = findViewById(R.id.txtSelected);
```

```
        CompoundButton.OnCheckedChangeListener listener = (buttonView, isChecked) ->  
        updateSelectedItems();
```

```
chkMilk.setOnCheckedChangeListener(listener);  
chkBread.setOnCheckedChangeListener(listener);  
chkEggs.setOnCheckedChangeListener(listener);  
chkRice.setOnCheckedChangeListener(listener);  
chkOil.setOnCheckedChangeListener(listener);  
}  
  
private void updateSelectedItems() {  
    StringBuilder selected = new StringBuilder("Selected Items:\n");  
  
    if (chkMilk.isChecked()) selected.append("• Milk\n");  
    if (chkBread.isChecked()) selected.append("• Bread\n");  
    if (chkEggs.isChecked()) selected.append("• Eggs\n");  
    if (chkRice.isChecked()) selected.append("• Rice\n");  
    if (chkOil.isChecked()) selected.append("• Oil\n");  
  
    if (selected.toString().equals("Selected Items:\n"))  
        txtSelected.setText("No items selected");  
    else  
        txtSelected.setText(selected.toString());  
}  
}
```

OUTPUT:



PRACTICAL NO: 10

AIM: To develop an Android application with: **Registration Page** → User enters username + password, **Login Page** → Verifies username and password, On successful login → Show **“Welcome User”** Toast message

CODE SNIPPET:

activity_main.xml (Login Page)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/edtUsername"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Username" />

    <EditText
        android:id="@+id/edtPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Password"
        android:inputType="textPassword"
        android:layout_marginTop="10dp" />

    <Button
        android:id="@+id/btnLogin"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login"
        android:layout_marginTop="20dp" />

    <Button
        android:id="@+id/btnRegister"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Go to Registration"
        android:layout_marginTop="10dp" />
</LinearLayout>
```

activity_register.xml (Registration Page)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/edtNewUser"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Create Username" />

    <EditText
        android:id="@+id/edtNewPass"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Create Password"
        android:inputType="textPassword"
        android:layout_marginTop="10dp" />

    <Button
        android:id="@+id/btnSubmit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Register"
        android:layout_marginTop="20dp" />

</LinearLayout>
```

MainActivity.java (Login Page Logic)

```
package com.example.loginapp;

import androidx.appcompat.app.AppCompatActivity;
```

```
import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    EditText edtUsername, edtPassword;
    Button btnLogin, btnRegister;

    SharedPreferences sharedPref;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        edtUsername = findViewById(R.id.edtUsername);
        edtPassword = findViewById(R.id.edtPassword);
        btnLogin = findViewById(R.id.btnLogin);
        btnRegister = findViewById(R.id.btnRegister);

        sharedPref = getSharedPreferences("UserData", MODE_PRIVATE);

        btnLogin.setOnClickListener(v -> {
            String user = edtUsername.getText().toString();
            String pass = edtPassword.getText().toString();

            String regUser = sharedPref.getString("username", "");
            String regPass = sharedPref.getString("password", "");

            if (user.equals(regUser) && pass.equals(regPass)) {
                Toast.makeText(this, "Welcome " + user, Toast.LENGTH_LONG).show();
            } else {
                Toast.makeText(this, "Invalid Username or Password", Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

```
}  
});
```

```
btnRegister.setOnClickListener(v -> {  
    Intent i = new Intent(MainActivity.this, RegisterActivity.class);  
    startActivity(i);  
});  
}  
}
```

RegisterActivity.java (Registration Logic)

```
package com.example.loginapp;  
  
import androidx.appcompat.app.AppCompatActivity;  
import android.content.SharedPreferences;  
import android.os.Bundle;  
import android.widget.Button;  
import android.widget.EditText;  
import android.widget.Toast;  
  
public class RegisterActivity extends AppCompatActivity {  
  
    EditText edtNewUser, edtNewPass;  
    Button btnSubmit;  
    SharedPreferences sharedPref;  
  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        setContentView(R.layout.activity_register);  
    }  
}
```



```
edtNewUser = findViewById(R.id.edtNewUser);
```

```
edtNewPass = findViewById(R.id.edtNewPass);
```

```
btnSubmit = findViewById(R.id.btnSubmit);
```

```
sharedPref = getSharedPreferences("UserData", MODE_PRIVATE);
```

```
btnSubmit.setOnClickListener(v -> {
```

```
    String newUser = edtNewUser.getText().toString();
```

```
    String newPass = edtNewPass.getText().toString();
```

```
    SharedPreferences.Editor editor = sharedPref.edit();
```

```
    editor.putString("username", newUser);
```

```
    editor.putString("password", newPass);
```

```
    editor.apply();
```

```
    Toast.makeText(this, "User Registered Successfully!",  
    Toast.LENGTH_SHORT).show();
```

```
    finish(); // return to login page
```

```
});
```

```
}
```

```
}
```

OUTPUT:

PRACTICAL NO: 11

AIM: To develop an Android application with a Login Screen that verifies username and password.

Upon successful login, the app should navigate to another activity displaying “Welcome User” and a Logout button.

When the user clicks Logout, a confirmation dialog should appear with OK and Cancel options:

- OK → Navigate back to Login Screen
- Cancel → Stay on current screen

CODE SNIPPET:

activity_login.xml (UI for Login Screen)

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:orientation="vertical"
```

```
    android:gravity="center"
```

```
    android:padding="20dp"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent">
```

```
<EditText
```

```
    android:id="@+id/etUsername"
```

```
    android:hint="Enter Username"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"/>
```

```
<EditText
```

```
    android:id="@+id/etPassword"
```

```
    android:hint="Enter Password"
```

```
    android:inputType="textPassword"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"
```

```
    android:marginTop="10dp"/>
```

```
<Button
```

```
    android:id="@+id/btnLogin"
```

```
    android:text="Login"
```



```
android:layout_width="match_parent"  
android:layout_height="wrap_content"  
android:marginTop="20dp"/>
```

```
</LinearLayout>
```

LoginActivity.java

```
package com.example.loginapp;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.content.Intent;
```

```
import android.os.Bundle;
```

```
import android.view.View;
```

```
import android.widget.Button;
```

```
import android.widget.EditText;
```

```
import android.widget.Toast;
```

```
public class LoginActivity extends AppCompatActivity {
```

```
    EditText etUsername, etPassword;
```

```
    Button btnLogin;
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_login);
```

```
        etUsername = findViewById(R.id.etUsername);
```

```
        etPassword = findViewById(R.id.etPassword);
```

```
        btnLogin = findViewById(R.id.btnLogin);
```

```
        btnLogin.setOnClickListener(new View.OnClickListener() {
```

```
            @Override
```

```
            public void onClick(View v) {
```

```
                String user = etUsername.getText().toString();
```

```
                String pass = etPassword.getText().toString();
```

```
// Simple username/password check
if (user.equals("admin") && pass.equals("1234")) {

    Intent i = new Intent(LoginActivity.this, WelcomeActivity.class);
    i.putExtra("username", user);
    startActivity(i);
    finish(); // Prevents going back with Back button

} else {
    Toast.makeText(LoginActivity.this, "Invalid Credentials!",
Toast.LENGTH_SHORT).show();
}
}
});
}
}
```

activity_welcome.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:orientation="vertical"

    android:gravity="center"

    android:padding="20dp"

    android:layout_width="match_parent"

    android:layout_height="match_parent">

    <TextView

        android:id="@+id/tvWelcome"

        android:textSize="24sp"

        android:text="Welcome User"

        android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"/>
```

```
<Button
```

```
    android:id="@+id/btnLogout"
```

```
    android:text="Logout"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"
```

```
    android:marginTop="30dp"/>
```

```
</LinearLayout>
```

WelcomeActivity.java

```
package com.example.loginapp;
```

```
import androidx.appcompat.app.AlertDialog;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.content.DialogInterface;
```

```
import android.content.Intent;
```

```
import android.os.Bundle;
```

```
import android.view.View;
```

```
import android.widget.Button;
```

```
import android.widget.TextView;
```

```
public class WelcomeActivity extends AppCompatActivity {
```

```
    TextView tvWelcome;
```

```
    Button btnLogout;
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_welcome);
```

```
        tvWelcome = findViewById(R.id.tvWelcome);
```

```
btnLogout = findViewById(R.id.btnLogout);

// Get username from intent
String username = getIntent().getStringExtra("username");
tvWelcome.setText("Welcome " + username);

btnLogout.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

        // Confirmation Dialog
        AlertDialog.Builder builder = new AlertDialog.Builder(WelcomeActivity.this);
        builder.setTitle("Logout Confirmation");
        builder.setMessage("Are you sure you want to logout?");

        builder.setPositiveButton("OK", new DialogInterface.OnClickListener() {
            @Override
            public void onClick(DialogInterface dialog, int which) {

                Intent i = new Intent(WelcomeActivity.this, LoginActivity.class);
                startActivity(i);
                finish();
            }
        });

        builder.setNegativeButton("Cancel", null);

        builder.show();
    }
});
}
```

OUTPUT:



PRACTICAL NO: 12

AIM: To create an Android app that contains a **menu with five options**.

When the user selects any menu option, the **selected option should be displayed in a TextView** on the screen.

CODE SNIPPET:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:id="@+id/tvResult"
        android:text="Selected: None"
        android:textSize="22sp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

menu_main.xml (Menu file inside res/menu/ folder)

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item android:id="@+id/option1" android:title="Option 1"/>
    <item android:id="@+id/option2" android:title="Option 2"/>
    <item android:id="@+id/option3" android:title="Option 3"/>
    <item android:id="@+id/option4" android:title="Option 4"/>

</menu>
```

```
<item android:id="@+id/option5" android:title="Option 5"/>
```

```
</menu>
```

MainActivity.java

```
package com.example.menuapp;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.os.Bundle;
```

```
import android.view.Menu;
```

```
import android.view.MenuItem;
```

```
import android.widget.TextView;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    TextView tvResult;
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);
```

```
        tvResult = findViewById(R.id.tvResult);
```

```
    }
```

```
    @Override
```

```
    public boolean onCreateOptionsMenu(Menu menu) {
```

```
        getMenuInflater().inflate(R.menu.menu_main, menu);
```

```
        return true;
```

```
    }
```

```
    @Override
```

```
    public boolean onOptionsItemSelected(MenuItem item) {
```

```
        switch (item.getItemId()) {
```

```
case R.id.option1:
    tvResult.setText("Selected: Option 1");
    return true;

case R.id.option2:
    tvResult.setText("Selected: Option 2");
    return true;

case R.id.option3:
    tvResult.setText("Selected: Option 3");
    return true;

case R.id.option4:
    tvResult.setText("Selected: Option 4");
    return true;

case R.id.option5:
    tvResult.setText("Selected: Option 5");
    return true;
}

return super.onOptionsItemSelected(item);
}
}
```

OUTPUT:



PRACTICAL NO: 13

AIM: TO Build an app using LinearLayout: Create a simple app that uses LinearLayout. It should take the contents of a predefined TextView, convert it to uppercase on button click, and display it in an EditText. Additionally, create an app that responds to key events in the EditText without needing a button press.

CODE SNIPPET:

activity_main.xml (LinearLayout usage)

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:orientation="vertical"

    android:padding="20dp"

    android:gravity="center"

    android:layout_width="match_parent"

    android:layout_height="match_parent">

    <TextView

        android:id="@+id/tvSource"

        android:text="hello android programming"

        android:textSize="22sp"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"/>

    <EditText

        android:id="@+id/etResult"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"
```



```
android:layout_marginTop="20dp"
```

```
android:hint="Converted Text"/>
```

```
<Button
```

```
    android:id="@+id/btnConvert"
```

```
    android:text="Convert to UPPERCASE"
```

```
    android:layout_marginTop="20dp"
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"/>
```

```
</LinearLayout>
```

MainActivity.java (Convert to UPPERCASE)

```
package com.example.linearlayoutapp;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.os.Bundle;
```

```
import android.view.View;
```

```
import android.widget.Button;
```

```
import android.widget.EditText;
```

```
import android.widget.TextView;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    TextView tvSource;
```

```
    EditText etResult;
```

```
    Button btnConvert;
```

```
@Override
```

```
protected void onCreate(Bundle savedInstanceState) {
```

```
    super.onCreate(savedInstanceState);
```

```
    setContentView(R.layout.activity_main);
```

```
tvSource = findViewById(R.id.tvSource);  
etResult = findViewById(R.id.etResult);  
btnConvert = findViewById(R.id.btnConvert);  
  
btnConvert.setOnClickListener(new View.OnClickListener() {  
    @Override  
    public void onClick(View v) {  
        String text = tvSource.getText().toString();  
        etResult.setText(text.toUpperCase());  
    }  
});  
}
```

OUTPUT:



PRACTICAL NO: 14

AIM: To Create an app with TableLayout and custom styles: Use a TableLayout with a TextView, EditText, and buttons. Also, create a custom styles.xml in the res/values directory to style the TextView.

CODE SNIPPET:

styles.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>

    <!-- Custom TextView Style -->
    <style name="CustomTableTextStyle">
        <item name="android:textSize">24sp</item>
        <item name="android:textColor">#FFFFFF</item>
        <item name="android:background">#3F51B5</item>
        <item name="android:padding">12dp</item>
        <item name="android:gravity">center</item>
        <item name="android:textStyle">bold</item>
    </style>

</resources>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<TableLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:padding="20dp"
    android:stretchColumns="1"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <!-- Row 1 -->
```

```
<TableRow>
<TextView
android:id="@+id/tvTitle"
style="@style/CustomTableTextStyle"
android:text="Enter Your Name"
android:layout_width="wrap_content"
android:layout_height="wrap_content"/>
</TableRow>
```

<!-- Row 2 -->

```
<TableRow>
<EditText
android:id="@+id/etName"
android:hint="Type here..."
android:layout_width="match_parent"
android:layout_height="wrap_content" />
</TableRow>
```

<!-- Row 3 -->

```
<TableRow>
<Button
android:id="@+id/btnShow"
android:text="Show"
android:layout_marginRight="10dp"
android:layout_width="wrap_content"
android:layout_height="wrap_content"/>
```

```
<Button
android:id="@+id/btnClear"
android:text="Clear"
android:layout_width="wrap_content"
android:layout_height="wrap_content"/>
</TableRow>
```

```
</TableLayout>
```

MainActivity.java

```
package com.example.tablelayoutapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    EditText etName;
    TextView tvTitle;
    Button btnShow, btnClear;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etName = findViewById(R.id.etName);
        tvTitle = findViewById(R.id.tvTitle);
        btnShow = findViewById(R.id.btnShow);
        btnClear = findViewById(R.id.btnClear);

        btnShow.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String name = etName.getText().toString();
                tvTitle.setText("Hello, " + name);
            }
        });

        btnClear.setOnClickListener(new View.OnClickListener() {
```

```
@Override  
public void onClick(View v) {  
    etName.setText("");  
    tvTitle.setText("Enter Your Name");  
}  
});  
}  
}
```

OUTPUT:

PRACTICAL NO: 15

AIM: To Create an app with SQLite database operations: Build an app that allows the user to perform CRUD operations (Create, Read, Update, Delete) with an SQLite database.

CODE SNIPPET:

DatabaseHelper.java

```
package com.example.sqlitecrud;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DatabaseHelper extends SQLiteOpenHelper {

    public static final String DB_NAME = "StudentDB";
    public static final String TABLE_NAME = "students";
    public static final String COL_1 = "ID";
    public static final String COL_2 = "NAME";
    public static final String COL_3 = "CITY";

    public DatabaseHelper(Context context) {
        super(context, DB_NAME, null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("CREATE TABLE " + TABLE_NAME +
            " (ID INTEGER PRIMARY KEY AUTOINCREMENT, NAME TEXT, CITY TEXT)");
    }
}
```

@Override

```
public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {  
    db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);  
    onCreate(db);  
}
```

// Insert Data

```
public boolean insertData(String name, String city) {  
    SQLiteDatabase db = this.getWritableDatabase();  
    ContentValues values = new ContentValues();  
    values.put(COL_2, name);  
    values.put(COL_3, city);  
    long result = db.insert(TABLE_NAME, null, values);  
    return result != -1;  
}
```

// Get All Data

```
public Cursor getAllData() {  
    SQLiteDatabase db = this.getWritableDatabase();  
    return db.rawQuery("SELECT * FROM " + TABLE_NAME, null);  
}
```

// Update Data

```
public boolean updateData(String id, String name, String city) {  
    SQLiteDatabase db = this.getWritableDatabase();  
    ContentValues values = new ContentValues();  
    values.put(COL_2, name);  
    values.put(COL_3, city);  
  
    int result = db.update(TABLE_NAME, values, "ID=?", new String[]{id});  
    return result > 0;  
}
```

// Delete Data

```
public Integer deleteData(String id) {  
    SQLiteDatabase db = this.getWritableDatabase();  
    return db.delete(TABLE_NAME, "ID=?", new String[]{id});  
}
```



```
}  
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>  
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent">  
  
    <LinearLayout  
        android:orientation="vertical"  
        android:padding="20dp"  
        android:gravity="center"  
        android:layout_width="match_parent"  
        android:layout_height="wrap_content">  
  
        <EditText  
            android:id="@+id/etId"  
            android:hint="Enter ID"  
            android:layout_width="match_parent"  
            android:layout_height="wrap_content" />  
  
        <EditText  
            android:id="@+id/etName"  
            android:hint="Enter Name"  
            android:layout_width="match_parent"  
            android:layout_height="wrap_content"  
            android:layout_marginTop="10dp"/>  
  
        <EditText  
            android:id="@+id/etCity"  
            android:hint="Enter City"  
            android:layout_width="match_parent"  
            android:layout_height="wrap_content"  
            android:layout_marginTop="10dp"/>
```

```
<Button
    android:id="@+id/btnInsert"
    android:text="Insert"
    android:layout_marginTop="20dp"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
<Button
    android:id="@+id/btnView"
    android:text="View All"
    android:layout_marginTop="10dp"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
<Button
    android:id="@+id/btnUpdate"
    android:text="Update"
    android:layout_marginTop="10dp"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
<Button
    android:id="@+id/btnDelete"
    android:text="Delete"
    android:layout_marginTop="10dp"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
</LinearLayout>
```

```
</ScrollView>
```

MainActivity.java

```
package com.example.sqlitecrud;
```

```
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.database.Cursor;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    DatabaseHelper myDb;
    EditText etId, etName, etCity;
    Button btnInsert, btnView, btnUpdate, btnDelete;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        myDb = new DatabaseHelper(this);

        etId = findViewById(R.id.etId);
        etName = findViewById(R.id.etName);
        etCity = findViewById(R.id.etCity);

        btnInsert = findViewById(R.id.btnInsert);
        btnView = findViewById(R.id.btnView);
        btnUpdate = findViewById(R.id.btnUpdate);
        btnDelete = findViewById(R.id.btnDelete);

        // Insert
        btnInsert.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                boolean inserted = myDb.insertData(
                    etName.getText().toString(),
                    etCity.getText().toString()
                );
            }
        });
    }
}
```

```
        Toast.makeText(MainActivity.this,
            inserted ? "Data Inserted" : "Insert Failed",
            Toast.LENGTH_SHORT).show();
    }
});

// View All
btnView.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        Cursor res = myDb.getAllData();
        if (res.getCount() == 0) {
            showMessage("Error", "No Data Found");
            return;
        }

        StringBuilder buffer = new StringBuilder();
        while (res.moveToNext()) {
            buffer.append("ID: ").append(res.getString(0)).append("\n");
            buffer.append("Name: ").append(res.getString(1)).append("\n");
            buffer.append("City: ").append(res.getString(2)).append("\n\n");
        }

        showMessage("Student Data", buffer.toString());
    }
});

// Update
btnUpdate.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        boolean updated = myDb.updateData(
            etId.getText().toString(),
            etName.getText().toString(),
            etCity.getText().toString()
        );
    }
});
```

```
        Toast.makeText(MainActivity.this,
                        updated ? "Data Updated" : "Update Failed",
                        Toast.LENGTH_SHORT).show();
    }
});

// Delete
btnDelete.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        Integer deleted = myDb.deleteData(etId.getText().toString());

        Toast.makeText(MainActivity.this,
                        deleted > 0 ? "Data Deleted" : "Delete Failed",
                        Toast.LENGTH_SHORT).show();
    }
});
}
```

```
// Show dialog
public void showMessage(String title, String message) {
    AlertDialog.Builder builder = new AlertDialog.Builder(this);
    builder.setCancelable(true);
    builder.setTitle(title);
    builder.setMessage(message);
    builder.show();
}
}
```

OUTPUT:

PRACTICAL NO: 16

AIM: To Create an app with three vertically aligned buttons: Develop an app with three buttons arranged vertically. When any button is selected, the screen color should change accordingly.

CODE SNIPPET:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:id="@+id/mainLayout">

    <Button
        android:id="@+id/btnRed"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Red"
        android:layout_margin="15dp"/>

    <Button
        android:id="@+id/btnGreen"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Green"
        android:layout_margin="15dp"/>

    <Button
        android:id="@+id/btnBlue"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Blue"
        android:layout_margin="15dp"/>
```

</LinearLayout>

MainActivity.java

```
package com.example.colorchanger;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.graphics.Color;
import android.view.View;
import android.widget.Button;
import android.widget.LinearLayout;

public class MainActivity extends AppCompatActivity {

    LinearLayout mainLayout;
    Button btnRed, btnGreen, btnBlue;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        mainLayout = findViewById(R.id.mainLayout);
        btnRed = findViewById(R.id.btnRed);
        btnGreen = findViewById(R.id.btnGreen);
        btnBlue = findViewById(R.id.btnBlue);

        btnRed.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                mainLayout.setBackgroundColor(Color.RED);
            }
        });

        btnGreen.setOnClickListener(new View.OnClickListener() {
```

```
@Override
public void onClick(View v) {
    mainLayout.setBackgroundColor(Color.GREEN);
}

});

btnBlue.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        mainLayout.setBackgroundColor(Color.BLUE);
    }
});
}
```

OUTPUT: